ROPES
&
KNOTS

Job Performance Requirements

NFPA 1010

6.3.12 (B)

6.3.20

6.5.1

Note: All objectives in this section are testable.

Introduction

Subject: Ropes & Knots
Note: Parts of a Rope…Running, Working, Standing

1. The "Running" part of the rope is the end used for work such as hoisting, pulling or belaying.
2. The "Working" part of the rope is the end used to form the knot (Commonly referred to as the "Loose End" or "Bitter End").
3. The "Standing" part of the rope is between the Running and Working end.

Note: Elements of a Knot…Bight, Loop, Round Turn

1. A "Bight" is formed by bending the rope back on itself while keeping the sides parallel.
2. A "Loop" is formed by crossing the side of a Bight over the Standing part.
3. A "Round Turn" is formed by further bending one side of the Loop.

   Note: All knots are formed with one or more of these elements.
   After a Hitch or Knot is formed by the Working end, ensure the excess rope remaining (Referred to as the "Tail") is **a minimum of six inches** to be able to tie an Overhand Knot (Safety).

   Note: Knots may be formed by multiple procedures. We are assessing the knot and the urgency to tie it quickly, not the method it is formed.

Setup:

1. A **rope** shall be affixed to an elevated location at one end. The other end shall be utilized to tie tools for hoisting and hand knots.
2. A **second rope** shall be affixed to an elevated location at one end. The other end shall be left in a rope bag and not accessible. This rope must be utilized for tying tools in-line (in the middle of the rope).
3. A **third rope** shall be located on the ground for tying hand knots and tag lines. **The diameter shall be a different size or color than the rope used for hoisting** (Tag line(s) shall be a minimum of 10' feet long).
4. All knots shall be tied within 2:00 minutes, regardless of if tying a hand knot or a tool for hoisting.
5. Time begins when the student touches the tool or rope. Time ends when student informs the examiner skill/task is demonstrated.
6. **It is NOT necessary to present the knot or tool when demonstrated.**
7. **All knots shall be dressed upon completion (Loose knots are NOT functional).**
8. **Safety shall be tied and snug against the knot.**

Performance Objective

Subject: Ropes & Knots

Task: Overhand Knot (Safety)

Procedure: Confirm order with command

1. Form a Loop in the rope and pass the end through the Loop.

   Note: To secure Hitches and Knots, a Safety shall be applied upon completion when necessary.

2. Take the “Tail” end and wrap it around the Standing part, passing it through the Loop created to demonstrate the Safety.
3. The Safety must be snug against the Hitch or Knot, no more than 2” inches of space.

Performance Objective

Subject: Ropes & Knots

Task: Clove Hitch on a Bight around an object

Note: This is the knot used to tie the Ladder Halyard on closed systems.

Procedure: Confirm order with command

1. Form a large Bight.
2. Pass the Bight end around an object.
3. Cross over the Standing part of Bight and pass around the object a second time in the same direction.
4. Thread the Bight end under itself and pull the ends tight to form the Clove Hitch. The Working and Standing part of the Bight should be in opposite directions.
5. A Safety must be tied and snug against the Hitch.

Performance Objective

Subject: Ropes & Knots

Task: Clove Hitch in hand

Procedure: Confirm order with command

1. Form an overhand Loop in the left hand.
2. Form an identical overhand Loop in the right hand.
   or;
3. Cross your arms and grasp the rope.
4. Uncross your arms without losing your grip on the rope.
5. Place the left-hand Loop over the right-hand Loop.

Performance Objective

Subject: Ropes & Knots

Task: Clove Hitch around an object

Procedure: Confirm order with command

1. Pass the Working end around an object.
2. Cross over the Standing part and pass around the object a second time in the same direction.
3. Thread the end under itself and pull the ends tight to form the Clove Hitch. The Working and Standing part of the rope should be in opposite directions (Visualize traffic traveling both directions under the overpass).
4. A Safety must be tied and snug against the Hitch no more than 2” inches of space.

Performance Objective

Subject: Ropes & Knots

Task: Becket Bend or Sheet Bend

Note: Joining two (2) ropes of unequal/different diameter.

Procedure: Confirm order with command

1. Form a Bight with the larger diameter rope.
2. Pass the Working end of the smaller diameter rope through the Bight and around both sides (Standing parts) of Bight.
3. Thread the Working end of the smaller diameter rope under itself and around both sides of the Bight.
4. Pull the ends tight to form the Becket Bend.
5. A Safety must be tied with the Tail of both ropes and snug against the knot.

<u>Performance Objective</u>

<u>Subject:</u> Ropes & Knots

<u>Task:</u> Bowline

<u>Procedure:</u> Confirm order with command

1. Form an overhand Loop in the Standing part of the rope.
2. Pass the Working end up through the Loop (Visualize rabbit jumps out of the hole).
3. Pass the Working end over the Loop, under and around the Standing part (Visualize rabbit runs around the tree).
4. Pass the Working end down through the Loop (Visualize rabbit jumps in the hole).
5. The Tail can be tied inside or outside.
6. Pull the Tail tight and dress the knot to form the Bowline.
7. A Safety must be tied on the Loop and snug against “within 2” inches of” the knot.

<u>Performance Objectives</u>

<u>Subject:</u> Ropes & Knots

<u>Task:</u> Bowline around an object

<u>Procedure:</u> Confirm order with command

1. Form an overhand Loop in the Standing part of the rope.
2. Pass the Working end around the object.
3. Pass the Working end up through the Loop (Visualize rabbit jumps out of the hole).
4. Pass the Working end over the Loop, under and around the Standing part (Visualize rabbit runs around the tree).
5. Pass the Working end down through the Loop (Visualize rabbit jumps in the hole).
6. The Tail can be tied inside or outside.
7. Pull the Tail tight and dress the knot to form the Bowline.
8. A Safety must be tied on the Loop and snug against “within 2” inches of” the knot.

<u>Note:</u> If tying inline on an enclosed handle you would use a <u>Double Bowline on a bight</u>. This would be accomplished by making a large Bight in the rope and following steps (1) through (8) above.

Performance Objective

Subject: Ropes & Knots

Task: Figure Eight Stopper

Procedure: Confirm order with command

1. Form a Loop.
2. Pass the Working end under and around the Standing part.
3. Thread the Working end through the Loop.

Performance Objective

Subject: Ropes & Knots

Task: Figure Eight on a Bight

Procedure: Confirm order with command

1. Form a large Bight.
2. Form a Loop with the Bight.
3. Pass the Bight under and around the Standing part.
4. Thread the Bight through the Loop.
5. Pull on the Bight and the Standing part tight to form the knot.

Performance Objective

Subject: Ropes & Knots

Task: Figure Eight Around an Object or Follow Through

Procedure: Confirm order with command

1. Tie a loose Figure Eight.
2. Pass the Working end around an object.
3. Thread the Working end through the Figure Eight, following the entire knot in reverse. The Tail must be beside the Standing part.
4. Pull tight and snug to form the knot.

Note: If tying inline on an enclosed handle you would use a Double Figure eight on a bight. This would be accomplished by making a large Bight in the rope and following steps (1) through (4) above

Performance Objective

Subject: Ropes & Knots

Task: Figure Eight Bend

Note: Joining two (2) ropes of equal/same diameter.

Procedure: Confirm order with command

1. Place two ropes of equal diameter lying parallel, with one rope above the other. Ensure the Working ends are pointing in opposite directions.
2. Tie a loose Figure Eight in one of the ropes.
3. With the second rope, place the Working end beside the Tail of the first knot.
4. Thread the Working end through the Figure Eight, following the entire knot in reverse. Both Tails must be beside the Standing part of both ropes (Opposite ends).

Performance Objective

Subject: Ropes & Knots

Task: Tag Line (second rope)

Note: Utilized to control tools and equipment being hoisted.

Procedure: Confirm order with command

1. With a second rope, tie a Clove Hitch with a Safety near or at the bottom of the object…or…a Bowline can be utilized for objects with enclosed handles.
2. If tying the tool in-line (in the middle of the rope), the excess rope is the tag line. You may need to tie a Clove Hitch on a Bight at the bottom of the tool.

NOTE: Tag Line (second rope) must be a minimum of 10' Feet long to reach the second (2^{nd}) floor of a building.

Performance Objective

Subject: Ropes & Knots

Task: Prepare a Ladder for hoisting

Note: For hoisting - Rope should be between ladder and building.
For lowering - Rope should be on outside of ladder.

Procedure: Confirm order with command

1. Tie a large Loop, Figure Eight on a Bight.
2. Slip the Loop through the rungs $^2/_3$ the length of the ladder **(9th rung from the butt).**
3. Bring the Loop up and slip it over both tips of the ladder.
4. Tighten the Loop on the ladder by pulling the slack out. **The Figure Eight should be in the area between the hooks and the 11th rung.**
5. Attach a Tag Line with a safety on the bottom rung of the ladder, utilizing a second rope or tie the ladder in-line.

Note: These procedures and measurements pertain to utilizing a 14 foot roof ladder.

Performance Objective

Subject: Ropes & Knots

Task: Prepare any tool with an enclosed handle for hoisting

Procedure: Confirm order with command

1. For Hoisting, tie a Bowline with a Safety or Figure Eight Follow Through around the enclosed handle or opening.
2. Attach a Tag Line with a safety near the bottom of the tool, utilizing a second rope or in-line.

   Note: The Tag Line can be tied with a Bowline or Clove Hitch.

Saw: Place the Hoist Line/Knot on the top guard handle;
Place the Tag Line/Knot on the bottom/lower handle.

Fan: Place the Hoist Line/Knot on the top handle;
Place the Tag Line/Knot on the bottom of tool.
If tying the fan using 1 handle, then the Tag Line should be tied diagonally across from the top handle at the bottom of the fan.

3. **If tying these tools in-line, the following knots shall be utilized:**
   a. **Double Bowline on a Bight**
   b. **Double Figure Eight on a Bight**

Performance Objective

Subject: Ropes & Knots

Task: Prepare a Pick Head (Fire Axe) for hoisting

Procedure: Confirm order with command

1. With the handle of the axe toward the building, tie a Clove Hitch with a Safety, on the handle against the head of the axe.
2. Place a Half Hitch around the blade of the axe head.
3. Place a second (2^{nd}) Half Hitch on the handle approximately 4 inches from the end.
4. Attach a Tag Line with a safety near the bottom of the tool, above the Clove Hitch, utilizing a second rope or ties the tool in-line.

Performance Objective

Subject: Ropes & Knots

Task: Prepare a Pike Pole for hoisting

Procedure: Confirm order with command

1. With the head of the pike pole toward the building, tie a Clove Hitch with a Safety, on the handle ¾ the length of the pike pole from the head ($^1/_4$ from the end of the handle).
2. Place a Half Hitch midway between the Clove Hitch and the head of the pike pole.
3. Place a second (2nd) Half Hitch on the head of the pike pole near the hook.
4. Attach a Tag Line with a safety near the bottom of the tool, above the Clove Hitch, utilizing a second rope or ties the tool in-line.

Performance Objective

Subject: Ropes & Knots

Task: Prepare an Uncharged Hose Line for Hoisting

Procedure: Confirm order with command

1. Grasp the nozzle and fold it onto the hose creating approximately a 5-foot fold.
2. Ensure the bale is closed and laying against the hose.
3. Tie a Clove Hitch with a Safety around the tip of the nozzle and the hose it is folded against.
4. Place a Half Hitch directly on the coupling of the nozzle and hose, ensuring all the slack is removed and rope is taut.
5. Place a second (2^{nd}) Half Hitch around the folded hose approximately 12 inches from the end of the fold. This creates a handle to assist in grasping and lifting the hose into the building.
6. A Tag Line is NOT needed.